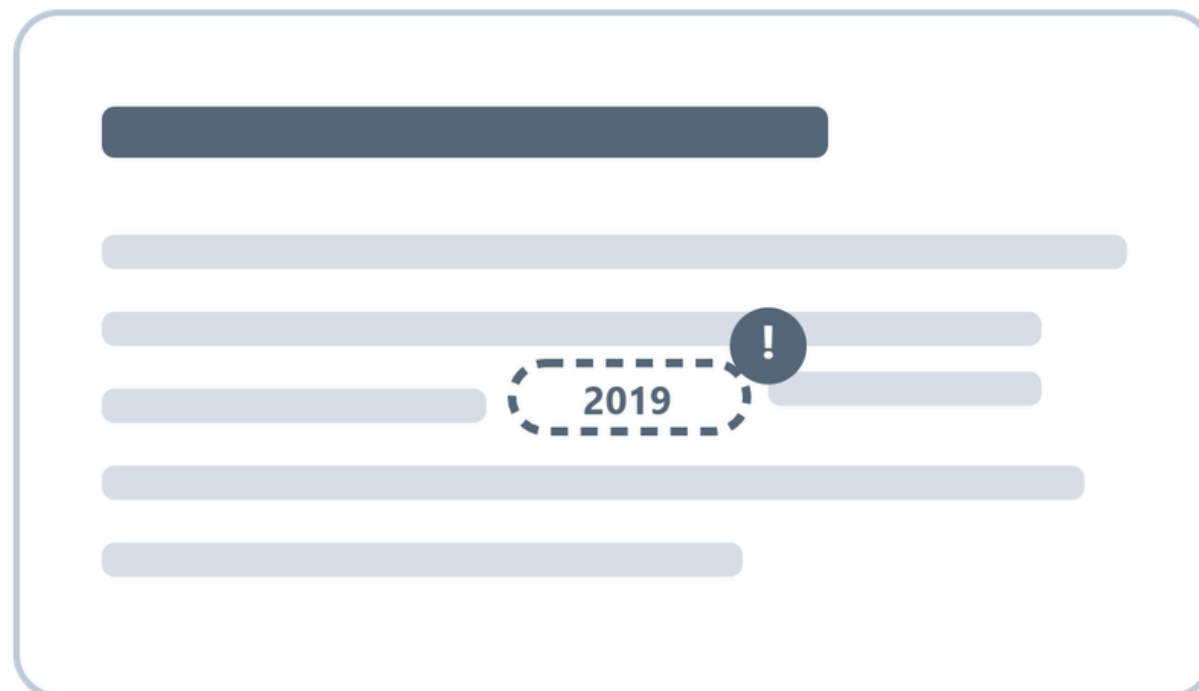

Detecting Temporal Inconsistencies in Political News through Temporal Knowledge Graphs

Pântea Marius-Nicușor
Bachelor's thesis, UTCN Cluj-Napoca

THE PROBLEM

A well-written lie

- Style-based detectors look at how something is written
- Correct language + wrong date = undetectable by style alone
- Example: a former official quoted as still holding office, years after leaving
- Easy to introduce, hard to catch just by reading

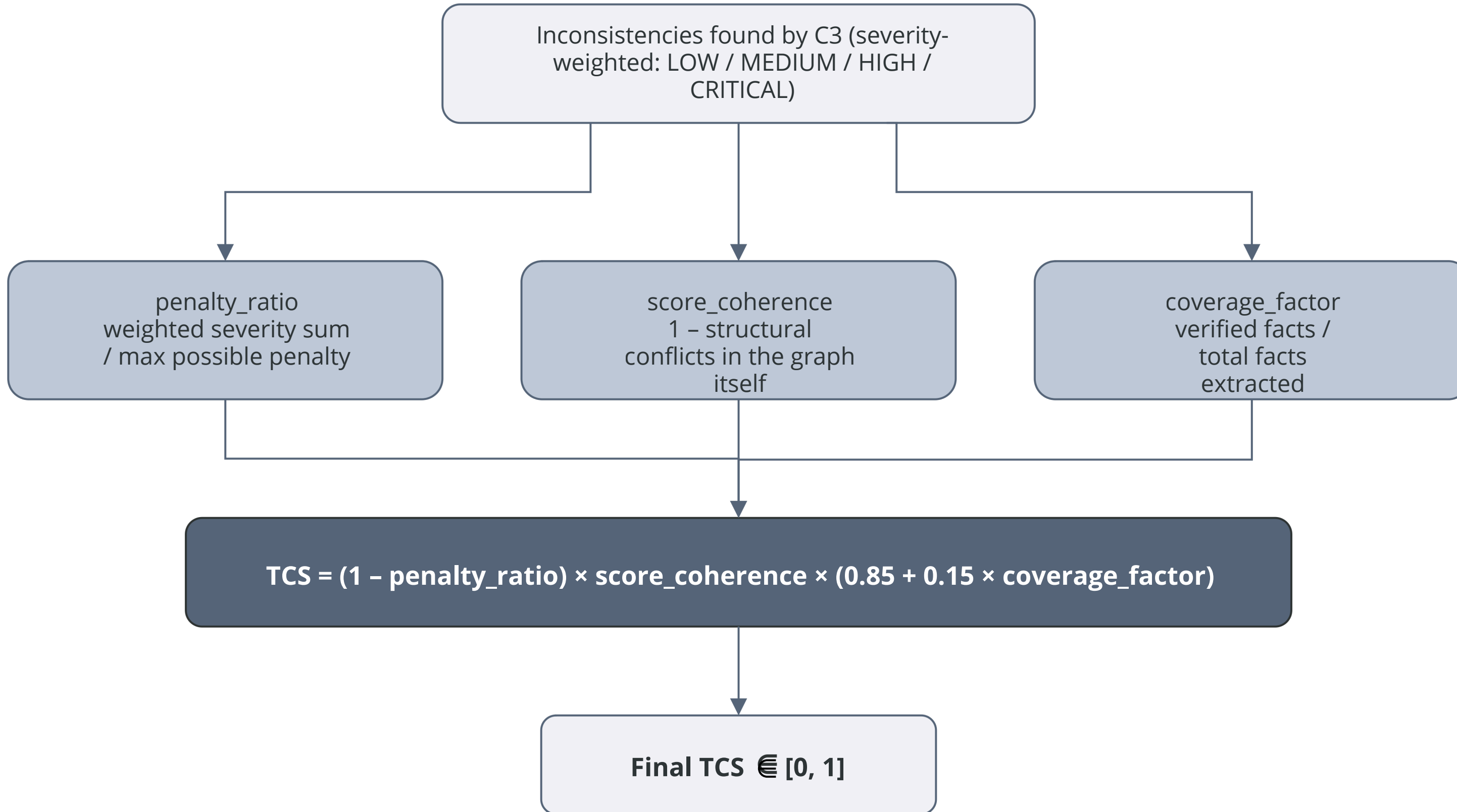


CONTRIBUTION: TCS

Temporal Coherence Score (TCS)

- Continuous score, $0 \rightarrow 1$, not a black box
- Three traceable factors: severity of inconsistencies, internal graph coherence, external verification coverage
- A verdict without a reason is nearly useless to a human reviewer — TCS always
- comes with the specific claim, entities, and contradicting source

TCS Score Breakdown



Example - coherent article

0 inconsistencies
penalty_ratio = 0
score_coherence = 1.0
coverage_factor = 0.9
→ TCS ≈ 0.99

Example - fake article

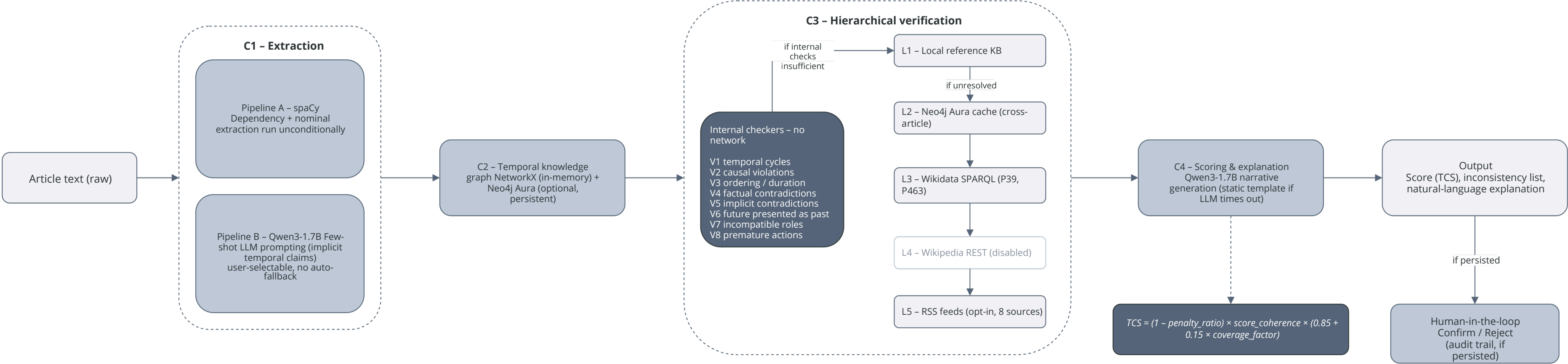
1 HIGH mandate-interval mismatch (V4)
penalty_ratio ≈ 0.67
score_coherence = 1.0
coverage_factor = 0.9
→ TCS ≈ 0.32

ARCHITECTURE

- Extraction — two user-selectable strategies (deterministic parsing / small LLM for implicit claims), not a fallback chain
- Temporal knowledge graph — structured, deduplicated representation
- Hierarchical verification — 8 internal checks (no network) running in parallel with a 5-level external escalation
- Scoring + explanation — TCS formula + natural-language justification



Architecture Diagram



RESULTS

Benchmark: 100 articles (56 true / 44 fake)

Metric	Value
Precision	0.909
Recall	0.227
F1	0.364
Accuracy	0.650

- Only 1 false alarm across the whole benchmark
- Tuned for precision: a false alarm costs more than a missed catch in this workflow
- Domain-specific: near-zero on general-purpose fake news datasets — expected, by design

LIMITATIONS

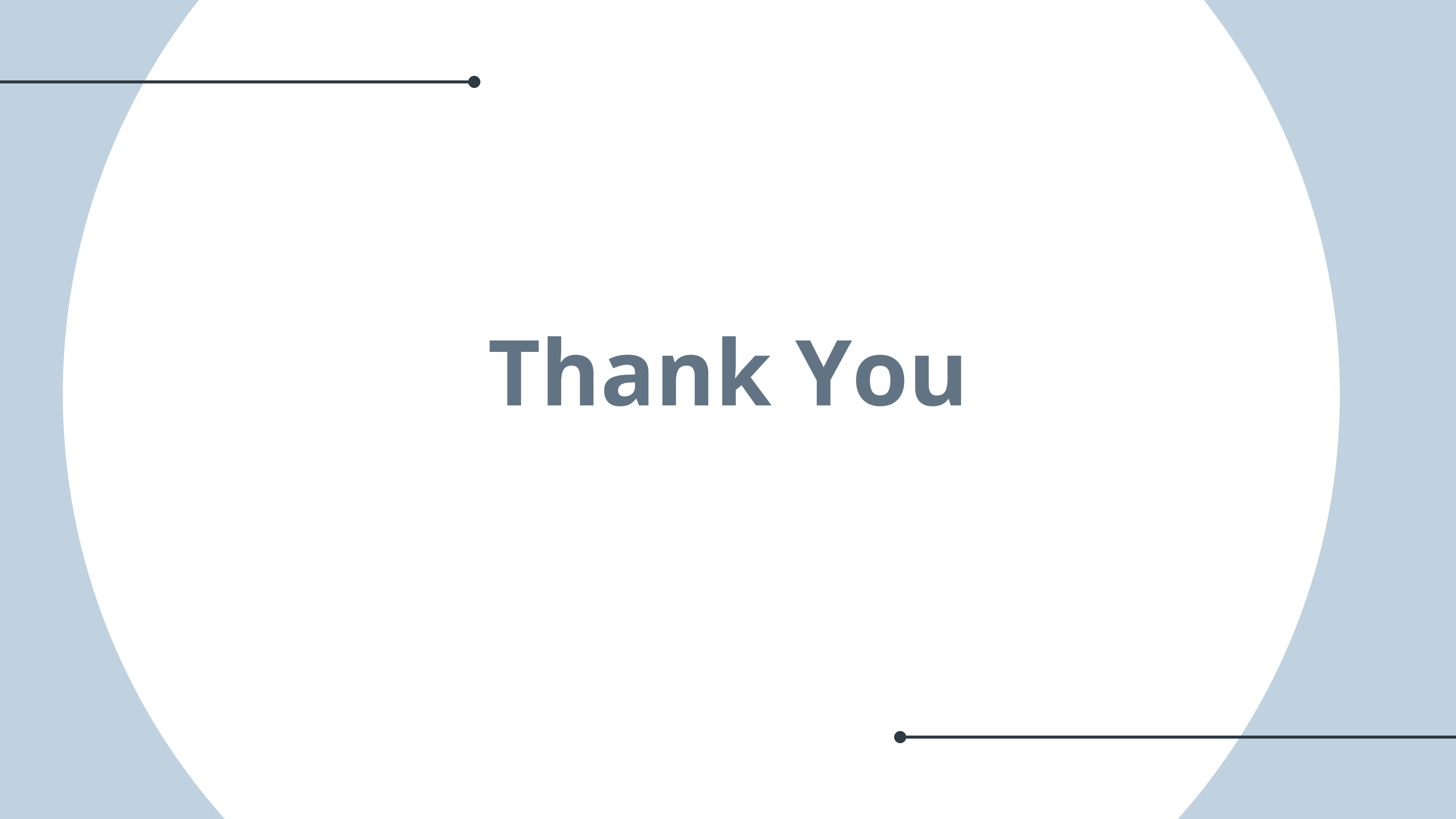
Where the system actually struggles

- Recall is low (0.227) — subtler or implicit errors often go undetected
- Benchmark is synthetic, manually annotated — not yet validated on real-world fake news
- Coverage is stronger for US/UK/EU figures than regional ones
- One known structural false positive: co-involvement in the same event vs. genuine contradiction — needs more narrative context to fully resolve

CONCLUSION

- Temporal coherence = a real, useful, explainable signal
- Complements existing detectors — doesn't replace them
- Built for the exact blind spot style-based methods miss — and it always shows its work





Thank You